

RAJPRAKASH SAHOO

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Summary

Full-stack engineer (2 years at product startups) now building AI systems end to end. **Shipped two production AI systems:** a multi-tenant LLM contract-diff platform (eval-gated, cost-accounted, human-in-the-loop) and a multimodal video search engine with measured retrieval quality and latency. Python, FastAPI, PostgreSQL; React/Next.js.

Projects

Shotgrep – Video Search Engine | *Python, FFmpeg, faster-whisper, SigLIP, LanceDB, FastAPI, MCP* [Live](#) | [GitHub](#)

- Built and shipped a multimodal video search engine: stage-cached, resumable ingest (probe, proxy, shot detection, ASR, frame embeddings) indexing a 60s 1080p clip in 72s on CUDA.
- Hybrid retrieval (SigLIP visual + dense/lexical transcript via rank fusion, shot-start priors, near-duplicate collapse) measured on 64 frozen queries over 3,797 moments: **Recall@5 0.688** vs 0.578 baseline, **MRR 0.495**, p50 189ms / p95 234ms. Served over REST and **MCP** with deep links into a Next.js player (API on Fly.io, web on Vercel).

Contract Change-Impact Intelligence | *FastAPI, pgvector, OpenAI, AWS, Terraform, Next.js* [Live](#) | [GitHub](#)

- Built and deployed a multi-tenant LLM platform that ingests agreements and amendments, extracts obligations, explains version diffs and maps each change to the clauses it impacts.
- Hallucination containment: every extraction must cite verifiable document text (citation gate, retry-then-drop) — **1.0 citation validity, 0.90 extraction precision**; hybrid retrieval (pgvector + full-text via Reciprocal Rank Fusion) at **0.96 recall@10**.
- PostgreSQL job queue (SKIP LOCKED, lease-based retries) replacing Celery/Redis; per-tenant LLM spend limits with per-call USD cost persisted; deployed on AWS (Terraform, ECS/Fargate, RDS, EFS) with Sentry and CloudWatch alarms; CI = ruff + strict mypy.

RN-OTP-TIMER | *React Native, npm* [Package](#) | [Github](#)

- standalone, fully customizable OTP timer npm package for React Native apps with **7,000+ downloads**.

Experience

Geeks Invention

Senior Software Engineer (Fullstack)

Remote
Aug 2022 – Feb 2024

- Built an appointment scheduler (Calendly replacement) handling core booking, cancellation and video-link integration backend logic with Next.js, Redux Toolkit and Firebase, driving **3x user growth**.
- Led OAuth, email verification and transactional email integration flows, lifting **user retention 20%**; tested with Cypress and React Testing Library.

Innovaccer

Software Developer 1 (Frontend)

Noida, India
Jan 2022 – Aug 2022

- Fixed 120+ previously failing unit tests and wrote 80+ new ones with React, Redux, Node.js, Python and Jest, raising frontend unit-test coverage from **10% to 75%**.
- Resolved 150+ bugs (70% of open frontend defects) to a zero-bug state within 3–4 months on client priority, earning the **best team award**.

UPSC Civil Services Preparation (career break) | *Self-directed full-time study; pivoted to AI engineering (May 2026)* | Apr 2024 – May 2026

Education

Bachelor of Technology in Computer Science | IIIT Bhubaneswar | 2018 – 2022

Technical Skills

Frontend: React.js, Next.js, TypeScript, Tailwind CSS, Redux Toolkit, TanStack Query, Cypress **Backend:** Python, FastAPI, Pydantic v2, Node.js, SQLAlchemy, PostgreSQL, Supabase **AI/LLM:** RAG, Evals (precision/recall, citation gates), Structured Outputs, pgvector, MCP, Multimodal (SigLIP/bge, ONNX int8), LanceDB, ASR (faster-whisper), LLM cost tracking, Agents **DevOps/Tools:** Git, Docker, Terraform, AWS (ECS/Fargate, RDS, EFS, S3), GitHub Actions, Fly.io